DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B. Tech **Branch : Computer Engineering** **Semester : IV**

Subject Code & Name: BTCOE406B_Y19 **Numerical Methods**

Max Marks: 60 **Date: 14 /07/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Students are required to attempt any four questions from Question No.2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**		**12**
1	The root of the equation $x^2 - 4x - 9 = 0$ using the bisection method up to four decimal places is a. 2.7065 b. 2.6875 c. 2.750 d. None	**CO-1**	1
2	Solve the following equation by Gauss Seidal Method up to 2 iterations and find the value of z. $27x + 6y - z = 85, 6x + 15y + 2z = 72,\ x + y + 54z = 85$ a. 0 b. 1.92 c. 1.88 d. 1.2	**CO-1**	1
3	In the Gauss elimination method for solving a system of linear algebraic equations, triangularization leads to a. Diagonal matrix b. Lower triangular matrix c. Upper triangular matrix d. Singular matrix	**CO-2**	1
4	In Gauss Jordan method which of the following transformations are allowed? a. Diagonal transformation b. Column transformation c. Row transformation d. square transformation	**C0-2**	1

5	Which of the following methods is capable of providing sufficient accuracy? a. Area by planimeter b. Area by co-ordinates c. Prismoid method d. Trapezoidal method	**CO-2**	1
6	Prismoid correction can be applied to the trapezoidal formula a. True b. False	**CO-2**	1
7	LU decomposition is used when a. $Ax(k)$=$y(k)$ b. A$x(k)$=$bx(k)$ c. $Ax(k)$≠$bx(k)$ d. $Ax(k)$=$b(k)$	**CO-1**	1
8	Find n if x_0 = 0.75825, x= 0.759 and h= 0.00005 a. 1.5 b. 15 c. 2.5 d. 25	**CO-1**	1
9	The velocity (m/s) of a body is given as a function of time (seconds) by v(t) = 200 ln (1+ t) − t, t > = 0 Using Euler's method with a step size of 5 seconds, the distance travelled in meters by the body from t = 2 to t = 12 seconds is most nearly a. 3133.1 b. 3939.7 c. 5698.0 d. 39397	**CO-1**	1
10	Find the values of x, y, z in the following system of equations by gauss Elimination Method. $2x + y - 3z = -10$, $-2y + z = -2$, $z = 6$ a. $x = 2$, $y = 4$, $z = 6$ b. $x = 2$, $y = 7$, $z = 6$ c. $x = 1$, $y = 4$, $z = 3$ d. $x = 2$, $y = 4$, $z = 5$	**CO-1**	1
11	The first two steps of the fourth-order Runge-Kutta method use _ a. Euler methods b. Forward Euler method c. Backward Euler method d. Explicit Euler method	**CO-2**	1
12	How many steps does the fourth-order Runge-Kutta method use? a. Two steps b. Three steps c. Four steps d. Five steps	**CO-2**	1

Q. 2 Solve the following. **12**

A) Find by false position method the real root of $x^3 - 4x - 9 = 0$ correct to three decimal places. **CO-1** **6**

B) Find a root of equation $x\sin x + \cos x = 0$ correct to four decimal places by using Newton-Raphson method **CO-1** **6**

Q.3 Solve the following. **12**

A) Solve the following system of equations by Gauss elimination method. **CO-1** **6**

$2x + y - z = 8,\ -3x - y + 2z = -11\ , -2x + y + 2z = -3$

B) Solve the following system of equations by Gauss Seidel iteration method. **CO-1** **6**

$8x - 3y + 2z = 20,$

$4x + 11y - z = 33,$

$6x + 3y + 12z = 35$

Q. 4 Solve Any Two of the following. **12**

A) The population of the city in the decimal sensor is given in the table. Find approx. population in 1925. **CO-1** **6**

Year	1891	1901	1911	1921	1931
Population	46	60	81	93	101

CO-1 **6**

B) Using Newton's backward interpolation formula. Find annual premium at the age of 33 from the following table.

Age in years	24	28	32	36	40
Premium in Rs.	28.06	30.19	32.75	34.94	40

CO-1 **6**

C) Find the value of 46 using the following data using Newton's forward interpolation formula.

Value	45	50	55	60	65
Approx.	114.84	96.16	83.22	74.48	68.48

Q.5 Solve Any Two of the following. **12**

A)	Evaluate $I = \int_0^1 \frac{dx}{1+x}$ using Trapezoidal rule.	**CO-1**	**6**
B)	Evaluate $I = \int_0^6 \frac{dx}{1+x^2}$ using Simpson's one third rule.	**CO-1**	**6**
C)	Evaluate $I = \int_0^1 \frac{2}{1+x^3}$ using Simpson's three eight rule.	**CO-1**	**6**

Q. 6 Solve Any Two of the following. **12**

A)	Using Taylor series method, correct to four decimal places, the value of y (0.1), given that $\frac{dy}{dx} = x^2 + y^2$ and y (0) = 1	**CO-1**	**6**
B)	Given $y'= -y$ and $y(0) = 1$**,** determine the value of y at $x = (0.01)(0.01)(0.04)$ by Euler's method.	**CO-1**	**6**
C)	Given $y'' + xy' + y = 0,\ y(0) = 1,\ y'(0) = 0,$ find the value of $y(0.1)$ by using Runge-Kutta method of fourth order.	**CO-1**	**6**

***** End *****